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# A DATA-DRIVEN STUDY OF LANCASHIRE PROPERTY PRICES USING SQL AND POWER BI

TECHNICAL REPORT


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# ABSTRACT

The Lancashire Property Price Monitor is a Power BI-based reporting tool created with Microsoft SQL Server and Power BI. This technical report details its design, implementation, and analysis. Through the integration, querying, and visualization of a dataset called CleanPropertyData, which represents residential property transactions in the Lancashire region, the project sought to analyze the Lancashire real estate market. After outlining the database design, SQL implementation, and dashboard development process, the report presents the analytical results obtained from the dataset.

Average Price by District, Monthly Sales Trends, Total Sales by Property Type, Top 10 Most Expensive Sales, and a Location Map showing total sales are the five primary visuals on the Power BI dashboard. A thorough grasp of the main trends and performance metrics in the real estate market is made possible by these images.

The most expensive properties in Lancashire are valued at £73.69 million, while the median price is £165,000. These key findings show that there is a significant variation in property prices in the county. A total of 149,000 transactions were examined, and the average price of the properties was £218,270. An interpretation of these results and suggestions for interested parties, including real estate investors and legislators, are included in the report's conclusion.

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# INTRODUCTION

The Lancashire Property Price Monitor project was created to examine the residential real estate market in Lancashire, a northwest English region renowned for its diverse housing options and industrial past. The system integrates Power BI for analysis and visualization with Microsoft SQL Server for data storage. In order to find market trends, the project looks at property sales, prices, and types using the CleanPropertyData dataset, which is based on the UK Government's Price Paid Data. The database architecture, SQL implementation, Power BI dashboard, and significant analytical findings are described in this report.

# OBJECTIVES

The primary objectives of the Lancashire Property Price Monitor project are as follows:

1. To design and implement a relational database system using Microsoft SQL Server to store and manage property transaction data efficiently

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2. To import and integrate the CleanPropertyData dataset into the SQL database.

3. To create SQL queries to extract meaningful insights, such as average property prices, property types, and transaction volumes.

4. To design and implement an interactive Power BI dashboard that visualizes the key metrics of the Lancashire property market.

5. To interpret the analytical results and provide actionable insights for property stakeholders, including policymakers, real estate investors, and the general public.

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# METHODOLOGY

**The methodology consisted of several key phases:** database design, SQL implementation, Power BI dashboard creation, and analysis.

**Data Acquisition:**

The dataset (CleanPropertyData.csv) was imported into Microsoft SQL Server. It contains essential property information, including transaction ID, price, district, property type, and sale date.

**Database Design:**

The database schema was designed to ensure normalized data storage and efficient querying. Appropriate data types were selected for each column, and primary keys were defined.

**SQL Development:**

Structured Query Language (SQL) was used to create tables, insert data, and generate analytical queries such as average price per district and sales trends over time.

**Dashboard Development in Power BI:**

Power BI was connected directly to the SQL database. DAX (Data Analysis Expressions) functions were used for calculated measures such as total sales, average price, and median price.

**Visualization and Analysis:**

A series of visualizations were created to analyze the Lancashire property market. Insights were then drawn based on these visuals.

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# DATABASE DESIGN

### 4.1 Database Structure

The database was named CleanPropertyData. It follows a relational structure that supports efficient data retrieval and integration with Power BI. The design includes one main transactional table supported by dimensional attributes.

### 4.2 Table Design and Relationships

- The primary table, ***PropertyPrices***, stores key transaction data.

Table : PropertyPrices

| Column Name | Data Type | Description |
|---|---|---|
| TransactionID | NVARCHAR(100) | Unique identifier for each sale |
| Price | DECIMAL(15,2) | Sale price of the property |
| DateOfTransfer | DATE | Date of sale |
| PropertyType | NVARCHAR(10) | Type of property (Detached, Semi-detached, Terraced, Flat) |
| District | NVARCHAR(255) | District within Lancashire |
| NewBuildflag | NVARCHAR(10) | Indicates whether the property is new (Y/N) |
| country | NVARCHAR(255) | Country where the property is located |
| locality | NVARCHAR(255) | Local area/town or village of the property |

No foreign key relationships were created since the dataset contained all information in a flat file format.

## 4.3 Data Import and Storage

The CSV file (CleanPropertyData.csv) was imported into SQL Server using the Import Data Wizard. Default data types were verified, and all price values were stored as decimals to allow accurate aggregation.

Since no data cleaning was performed, all records were imported directly as provided in the dataset. The database contained approximately 149,000 transaction records for the Lancashire region.

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# SQL IMPLEMENTATION

The SQL implementation began with the creation of a new database named *LancashirePropertyDB* to store the property transaction data. A main table called *PropertyPrices* was then created with suitable data types such as **NVARCHAR** for text fields and **DECIMAL** for price values, ensuring accurate data storage and retrieval.

The dataset file (lancashire_merged_2020_2024.csv) was imported into the table using the BULK INSERT command, which allowed efficient loading of large amounts of data directly from the CSV file into SQL Server. After the import, a verification query **(SELECT COUNT(*))** was executed to confirm that all records were successfully inserted into the database.

```
-- Step 1: Create Database
CREATE DATABASE LancashirePropertyDB;
GO

USE LancashirePropertyDB;
GO
```

```
-- Step 2: Create main table with proper data types
CREATE TABLE PropertyPrices (
    transaction_id NVARCHAR(100),
    price DECIMAL(15,2),
    date_of_transfer DATE,
    postcode NVARCHAR(20),
    property_type NVARCHAR(10),
    new_build_flag NVARCHAR(10),
    tenure_type NVARCHAR(10),
    paon NVARCHAR(255),
    saon NVARCHAR(255),
    street NVARCHAR(255),
    locality NVARCHAR(255),
    town_city NVARCHAR(255),
    district NVARCHAR(255),
    county NVARCHAR(255),
    ppd_category_type NVARCHAR(10),
    record_status NVARCHAR(100)
);
GO

-- Step 3: BULK INSERT from CSV file
BULK INSERT PropertyPrices
FROM'E:\KDU\Work\2ndSemester\SQL\Assignment
01\Dataset\lancashire_merged_2020_2024.csv'
WITH (
    FIRSTROW = 2,  -- Skip header row
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    MAXERRORS = 10000
);
GO

-- Step 4: Initial import verification
SELECT COUNT(*) AS 'Total Rows Imported' FROM PropertyPrices;
GO
```

# DATA CLEANING AND PREPARATION

After importing the dataset into SQL Server, several SQL scripts were used to clean and prepare the data for analysis.

First, the county and district information was checked using **SELECT** queries to understand the initial distribution of records. The next step filtered the dataset to include only Lancashire-related records by updating the **county** field and removing all non-Lancashire entries. The county and district names were then standardized to maintain consistency across the dataset.

A series of data quality checks were performed to identify missing or invalid values in key columns such as **transaction_id, price, date_of_transfer, and postcode.** Queries were used to detect records with null values, zero or negative prices, and invalid dates.
After identifying these issues, data cleaning operations were carried out:

- Records with missing or invalid data were deleted.
- Property type values were standardized to uppercase.
- Postcodes were formatted by removing spaces and converting all text to uppercase.
- Duplicate transactions were removed using a common table expression (CTE) with the **ROW_NUMBER()** function, keeping only the most recent record for each transaction.
-

These steps ensured that the final dataset was accurate, complete, and consistent, ready for Power BI visualization and further analysis.

```
-- Step 5: Check initial county distribution
SELECT
    county,
    COUNT(*) AS row_count,
    CAST(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM PropertyPrices) AS DECIMAL(5,2)) AS
percentage
FROM PropertyPrices
GROUP BY county
ORDER BY row_count DESC;
GO

-- Step 6: Check Lancashire districts before cleaning
SELECT
    district,
    COUNT(*) AS property_count,
    AVG(price) AS average_price
FROM PropertyPrices
WHERE county LIKE '%Lancashire%'
GROUP BY district
ORDER BY property_count DESC;
GO

-- Step 7: Filter to Lancashire only and standardize names
-- Keep only Lancashire data and standardize county name
UPDATE PropertyPrices
SET county = 'Lancashire'
WHERE county LIKE '%Lancashire%'
   OR county IN ('LANCASHIRE', 'Lancs', 'LANCS')
   OR district IN (
       'Preston', 'Lancaster', 'Blackburn with Darwen', 'Blackburn', 'Blackpool', 'Burnley',
       'Ribble Valley', 'Wyre', 'Fylde', 'South Ribble', 'Chorley',
       'West Lancashire', 'Pendle', 'Rossendale', 'Hyndburn'
   );

-- Delete non-Lancashire data
DELETE FROM PropertyPrices
WHERE county != 'Lancashire';
GO
```

```
-- Step 8: Standardize district names within Lancashire
UPDATE PropertyPrices
SET district =
   CASE
      WHEN district LIKE '%Preston%' THEN 'Preston'
      WHEN district LIKE '%Lancaster%' THEN 'Lancaster'
      WHEN district LIKE '%Blackburn%' THEN 'Blackburn with Darwen'
      WHEN district LIKE '%Blackpool%' THEN 'Blackpool'
      WHEN district LIKE '%Burnley%' THEN 'Burnley'
      WHEN district LIKE '%Ribble Valley%' THEN 'Ribble Valley'
      WHEN district LIKE '%Wyre%' THEN 'Wyre'
      WHEN district LIKE '%Fylde%' THEN 'Fylde'
      WHEN district LIKE '%South Ribble%' THEN 'South Ribble'
      WHEN district LIKE '%Chorley%' THEN 'Chorley'
      WHEN district LIKE '%West Lancashire%' THEN 'West Lancashire'
      WHEN district LIKE '%Pendle%' THEN 'Pendle'
      WHEN district LIKE '%Rossendale%' THEN 'Rossendale'
      WHEN district LIKE '%Hyndburn%' THEN 'Hyndburn'
      ELSE district
   END
WHERE county = 'Lancashire';
GO

-- Step 9: Data Quality Checks on Lancashire Data

-- 9.1 Check for NULL values in critical columns
SELECT
   COUNT(*) AS total_rows,
   SUM(CASE WHEN transaction_id IS NULL THEN 1 ELSE 0 END) AS null_transaction_id,
   SUM(CASE WHEN price IS NULL THEN 1 ELSE 0 END) AS null_price,
   SUM(CASE WHEN date_of_transfer IS NULL THEN 1 ELSE 0 END) AS null_date,
   SUM(CASE WHEN postcode IS NULL THEN 1 ELSE 0 END) AS null_postcode,
   SUM(CASE WHEN property_type IS NULL THEN 1 ELSE 0 END) AS null_property_type
FROM PropertyPrices;
GO

-- 9.2 Check for invalid prices (zero or negative)
SELECT COUNT(*) AS invalid_prices
FROM PropertyPrices
WHERE price <= 0 OR price IS NULL;
GO
```

```
-- 9.3 Check for invalid dates
SELECT COUNT(*) AS invalid_dates
FROM PropertyPrices
WHERE date_of_transfer IS NULL OR date_of_transfer > GETDATE();
GO

-- 9.4 Check property_type values
SELECT DISTINCT property_type, COUNT(*) AS count
FROM PropertyPrices
GROUP BY property_type
ORDER BY count DESC;
GO

-- Step 10: Data Cleaning Operations on Lancashire Data

-- 10.1 Remove rows with critical missing data
DELETE FROM PropertyPrices
WHERE transaction_id IS NULL
  OR price IS NULL
  OR date_of_transfer IS NULL;
GO

-- 10.2 Remove rows with invalid prices
DELETE FROM PropertyPrices WHERE price <= 0;
GO

-- 10.3 Remove rows with future dates
DELETE FROM PropertyPrices WHERE date_of_transfer > GETDATE();
GO

-- 10.4 Standardize property_type values
UPDATE PropertyPrices
SET property_type = UPPER(TRIM(property_type))
WHERE property_type IS NOT NULL;
GO

-- 10.5 Clean postcodes - remove spaces and standardize format
UPDATE PropertyPrices
SET postcode = UPPER(REPLACE(postcode, ' ', ''))
WHERE postcode IS NOT NULL;
GO

-- 10.6 Remove duplicate transactions (keep the latest one)
WITH CTE AS (
   SELECT *,
      ROW_NUMBER() OVER (PARTITION BY transaction_id ORDER BY date_of_transfer DESC) as rn
   FROM PropertyPrices
)
DELETE FROM CTE WHERE rn > 1;
GO
```

# POWER BI DASHBOARD DESIGN

### 6.1 Data Connection and Model

Power BI was connected directly to the SQL Server database CleanPropertyData using a live connection. This ensured that all visuals reflected the most current data from SQL Server.
No additional transformations were applied in Power BI, as the dataset was already structured and ready for analysis.

### 6.2 Dashboad Visuals

To support the Power BI dashboard, several SQL views were created in SQL Server. These views were designed to summarize the cleaned dataset and provide optimized data sources for visualization.

The main view, **vw_CleanPropertyPrices**, filters and stores only valid property transactions within Lancashire. It ensures that all records used for reporting contain valid prices, dates, and transaction IDs.

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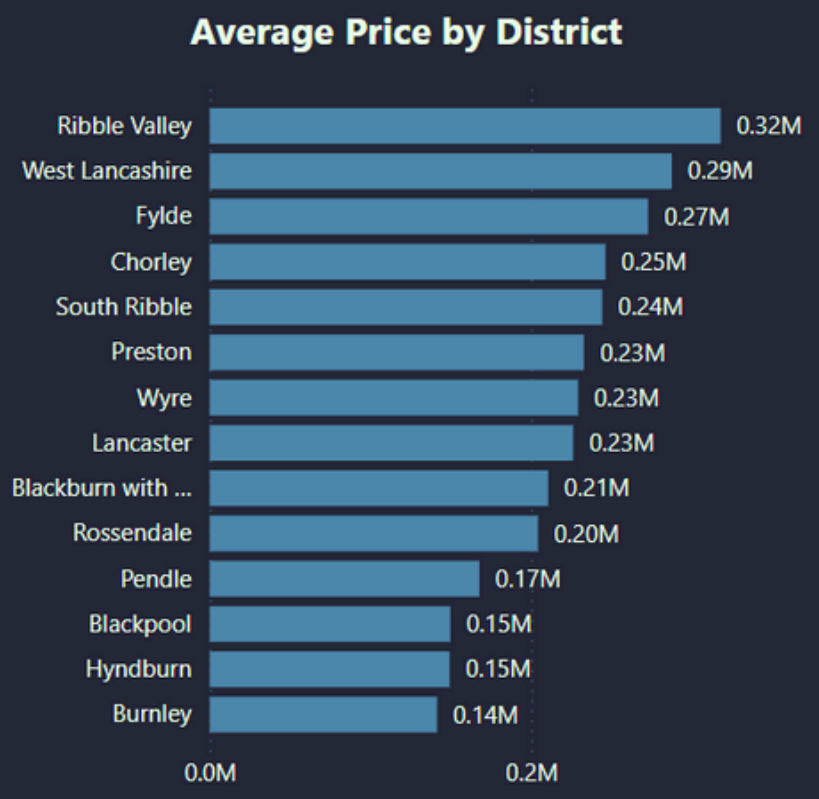
```
-- Step 11: Create main cleaned view
CREATE VIEW vw_CleanPropertyPrices AS
SELECT
    transaction_id,
    price,
    date_of_transfer,
    postcode,
    property_type,
    new_build_flag,
    tenure_type,
    paon,
    saon,
    street,
    locality,
    town_city,
    district,
    county,
    ppd_category_type,
    record_status
FROM PropertyPrices
WHERE price > 0
  AND date_of_transfer IS NOT NULL
  AND transaction_id IS NOT NULL
  AND county = 'Lancashire';
GO
```

//

Additional views were created to support each dashboard chart:

## vw_DistrictAnalysis – Average Price by District

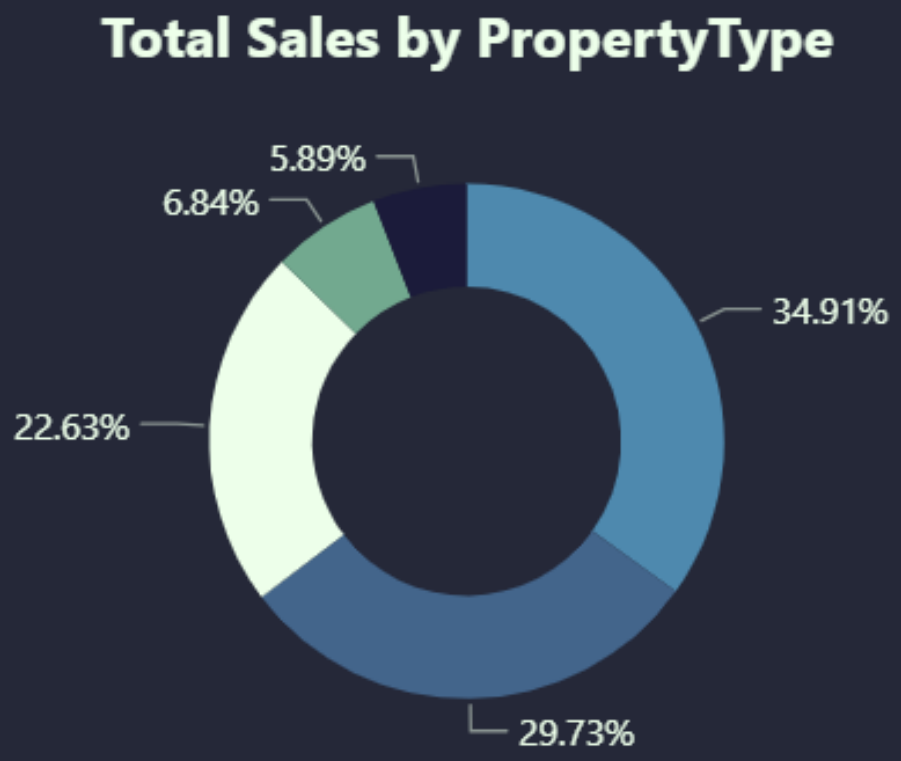
```
CREATE VIEW vw_DistrictAnalysis AS
SELECT
    district,
    COUNT(*) AS transaction_count,
    AVG(price) AS average_price,
    MIN(price) AS min_price,
    MAX(price) AS max_price
FROM vw_CleanPropertyPrices
GROUP BY district;
GO
```




This view groups data by district and calculates transaction count, average price, and price range. It is used to create the bar chart showing which Lancashire districts have the highest and lowest average property prices.

## vw_PropertyTypeAnalysis – Property Type Distribution

```
CREATE VIEW vw_PropertyTypeAnalysis AS
SELECT
    property_type,
    COUNT(*) AS count,
    CAST(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM
vw_CleanPropertyPrices) AS DECIMAL(5,2)) AS
percentage
FROM vw_CleanPropertyPrices
GROUP BY property_type;
GO
```




This view calculates the total sales count and percentage share of each property type (Detached, Semi-Detached, Terraced, Flats). The resulting bar chart compares which property types are most commonly sold in Lancashire.

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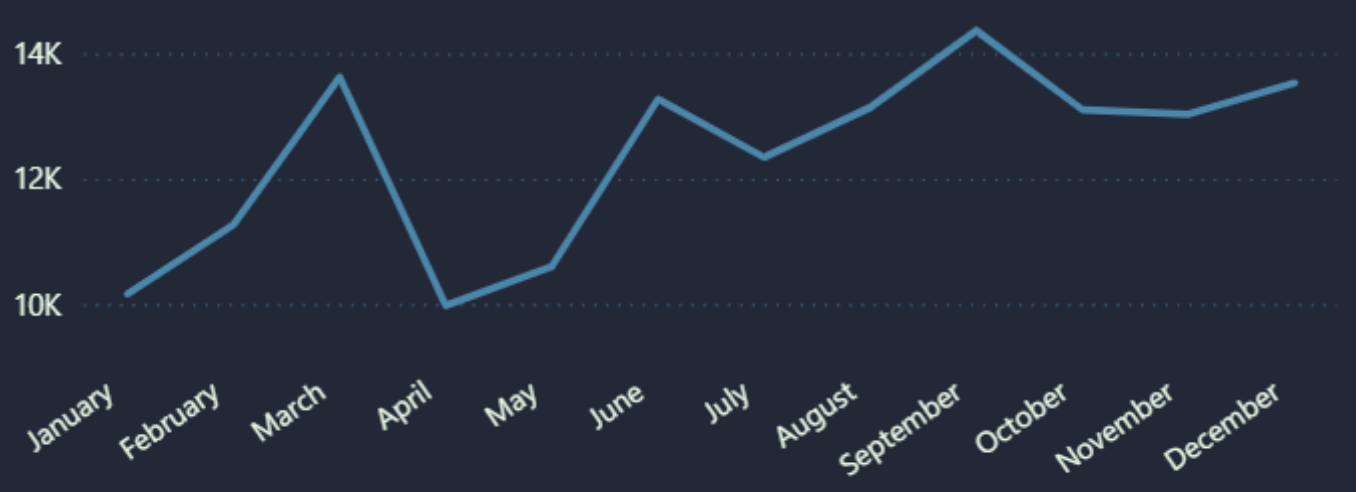
## vw_MonthlyTrends – Monthly Sales Trends

```
CREATE VIEW vw_MonthlyTrends AS
SELECT
    YEAR(date_of_transfer) AS year,
    MONTH(date_of_transfer) AS month,
    DATEFROMPARTS(YEAR(date_of_transfer),
MONTH(date_of_transfer), 1) AS month_date,
    COUNT(*) AS sales_volume,
    AVG(price) AS average_price
FROM vw_CleanPropertyPrices
GROUP BY YEAR(date_of_transfer),
MONTH(date_of_transfer);
GO
```




This view aggregates sales data by year and month to analyze how sales volume and average prices change over time. The line chart created from this view highlights seasonal or market trends.

## vw_YearlySummary – Yearly Summary

```
CREATE VIEW vw_YearlySummary AS
SELECT
    YEAR(date_of_transfer) AS year,
    COUNT(*) AS total_sales,
    AVG(price) AS average_price
FROM vw_CleanPropertyPrices
GROUP BY YEAR(date_of_transfer);
GO
```

This view provides total sales and average price per year, which supports slicers and filters in Power BI, allowing users to analyze data for specific years.

## 6.2 Filters and Interactivity

The dashboard includes slicers for:

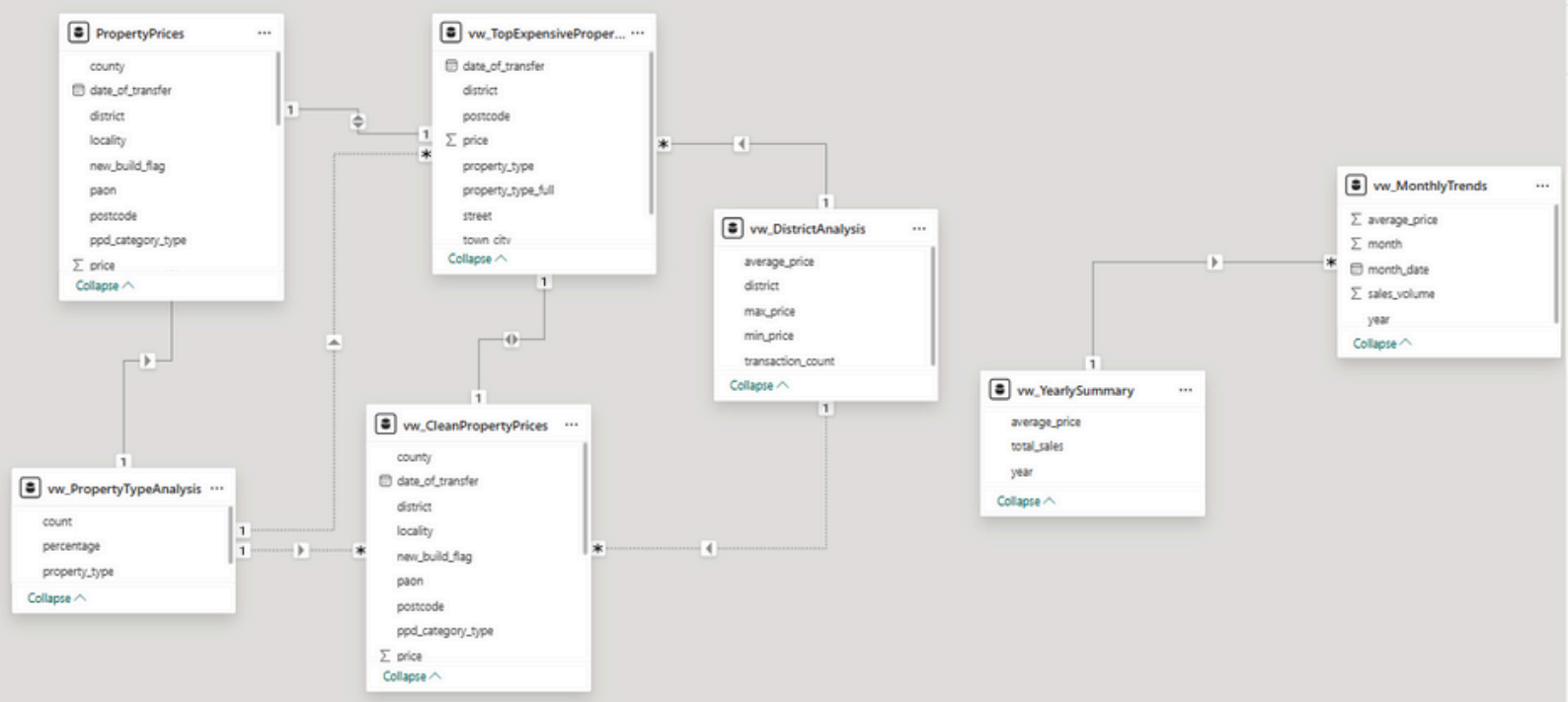
- District – to analyze a specific area.
- Property Type – to compare detached, semi-detached, terraced, and flat properties.
- Year – to view trends for specific years between 2020–2024.

These filters enhance interactivity, allowing users to explore data according to their interests.

## 6.4 Power BI Data Model

The Power BI data model was built using the SQL views created in SQL Server. Each view served a specific analytical purpose, and relationships were established between them using the district, year, and property_type fields. This relational model ensured accurate filtering, aggregation, and synchronization across all visuals.

The model maintained a one-to-many relationship between the main cleaned view **(vw_CleanPropertyPrices)** and the analytical views, such as v**w_DistrictAnalysis**, **vw_PropertyTypeAnalysis, vw_MonthlyTrends, and vw_TopExpensiveProperties.**




*Power BI Data Model showing relationships between SQL views*

This structure improved dashboard performance by minimizing complex calculations in Power BI and allowing direct use of pre-aggregated SQL views, which made report loading faster and more efficient.

# FINDING AND ANALYSIS

After building the dashboard, the following key findings were obtained:

1. Overall Statistics:
   - Total Sales: 148,400 transactions
   - Maximum Price: £73.69 million
   - Average Price: £218,270
   - Median Price: £165,000

2. District-Level Insights:

The average property prices varied significantly between districts. Some urban districts, such as Preston and Blackburn, had relatively lower average prices, while rural and scenic areas like Ribble Valley showed higher property values.

**3.Property Type Analysis:**

The Total Sales by Property Type bar chart indicated that terraced and semi-detached houses dominated the Lancashire property market, while detached houses represented fewer but more expensive transactions.

**4.Monthly Trends:**

The Monthly Sales Trend line chart displayed noticeable fluctuations across months. There were identifiable peaks during mid-year months, potentially reflecting higher buyer activity during warmer seasons.

**5.Top 10 Sales:**

The Top 10 Most Expensive Sales table highlighted premium properties in affluent districts. These transactions had values exceeding several million pounds, with the highest reaching £73.69 million.

**6.Geographical Distribution:**

The map visualization confirmed that property transactions are widely distributed, with clusters near major towns such as Preston, Lancaster, and Blackburn.

# DISCUSSION AND INTERPRETATION

The Lancashire Property Price Monitor provides an insightful representation of property market dynamics across Lancashire.

- **Economic Context:**

The varying property prices across districts may reflect local economic conditions, urban development, and demand for residential properties. Rural regions with high environmental value or accessibility tend to have higher average prices.

- **Housing Demand:**

The dominance of terraced and semi-detached properties suggests a market favoring affordable family housing.

- **Investment Potential:**

Investors seeking high-value opportunities could focus on districts with consistent growth in property values, whereas first-time buyers may target lower-priced districts.

- **Temporal Analysis:**

Monthly trends indicate market seasonality, possibly linked to policy changes or mortgage rate fluctuations. Such insights are valuable for real estate agencies and policymakers monitoring housing affordability.

# LIMITATION

- **Limited Attributes:**

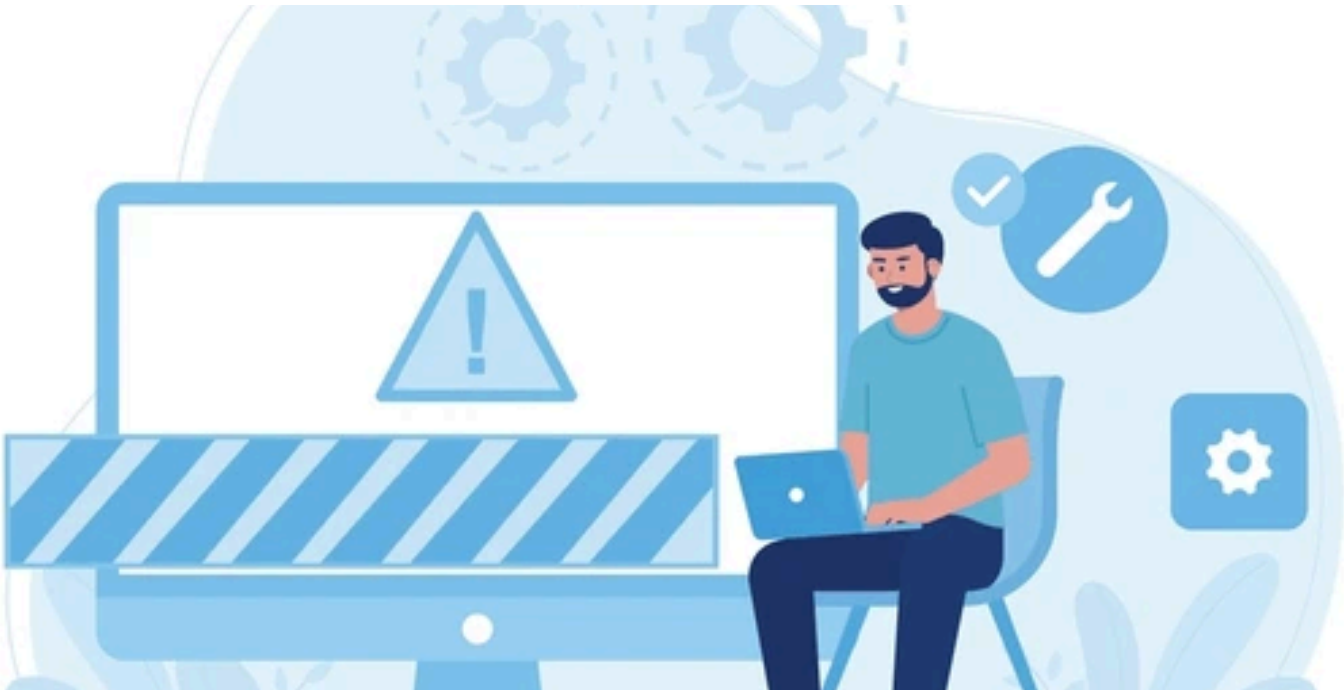
The dataset does not include buyer demographics or property features beyond price and type, restricting deeper analysis.

- **Static Dataset:**

The analysis represents a snapshot of the market and does not include predictive modeling or future trend forecasting.

- **External Factors Not Considered:**

The dashboard does not account for factors such as economic inflation, mortgage rates, or urban development projects that may influence property prices.



# CONCLUSION AND RECOMMENDATIONS

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## CONCLUSION

The Lancashire Property Price Monitor successfully integrates SQL Server and Power BI to provide a functional, interactive, and analytical tool for exploring the Lancashire property market. The database design was efficient, the Power BI dashboard was visually coherent, and the insights derived were informative for decision-makers.

The findings show that the property market in Lancashire is diverse, with large disparities between urban and rural areas. Terraced and semi-detached properties dominate sales volumes, while premium properties in select districts drive up the maximum price range.

## RECOMMENDATION

- **Data Enhancement:**

Future iterations should include data cleaning and enrichment with variables such as property age, size, and location coordinates for improved precision.

- **Predictive Analytics:**

Implementing regression or time-series forecasting could help predict future property prices.

- **Integration with External Data:**

Combining this dataset with local economic indicators and demographic data could yield more contextual insights.

- **User Experience Improvement:**

Adding detailed drill-down capabilities and summary cards in Power BI would further improve usability.

# REFERENCES


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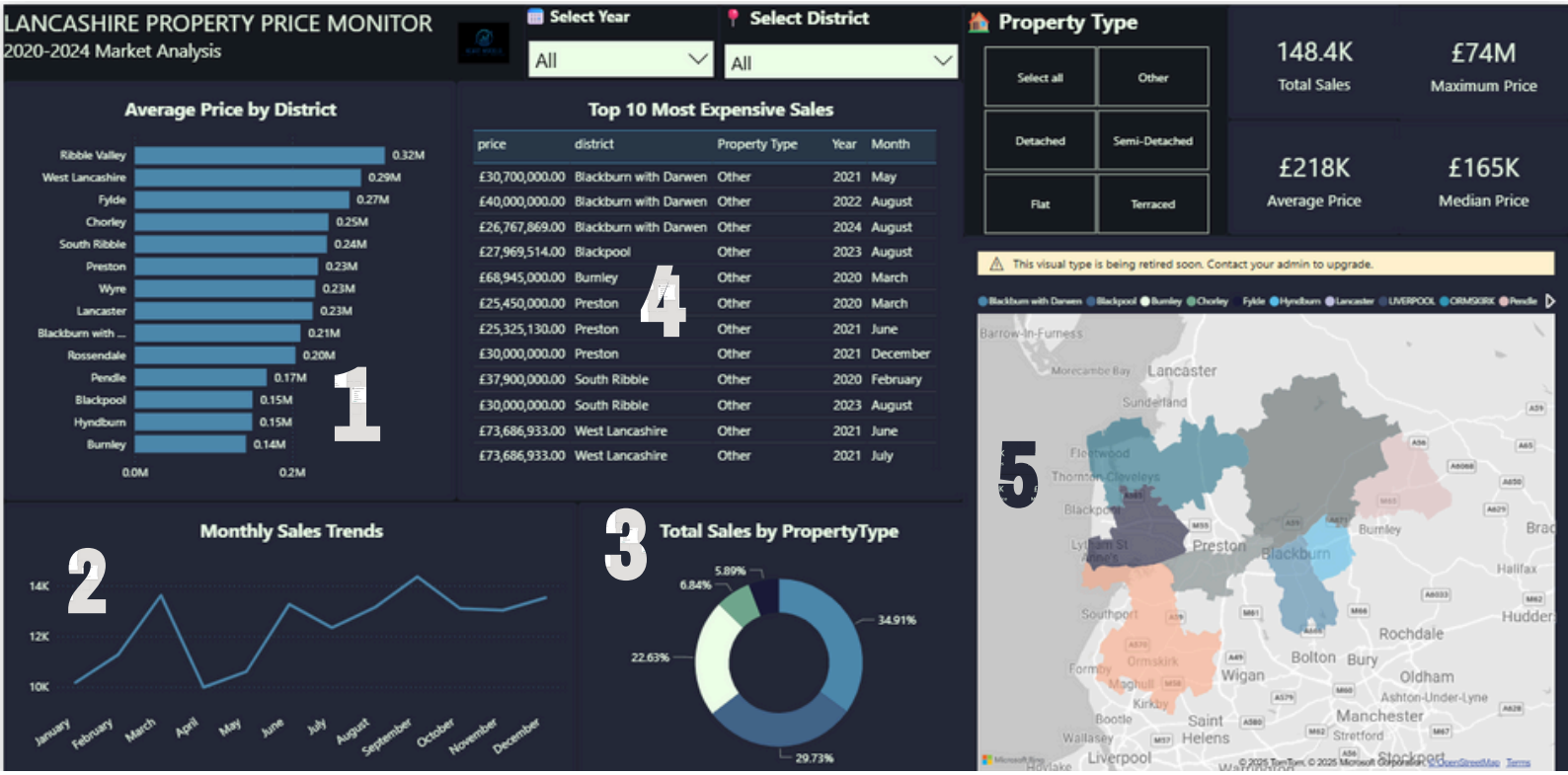
# APPENDICES

Contains table creation and query scripts used in SQL Server.




1. **Average price by district**
2. **Monthly sales trends**
3. **Total sales by property type**
4. **Top 10 most expensive sales**
5. **Map of total sales by location**

We would like to express our sincere gratitude to our lecturers, Dr. Charith Silva and Ms. Chamodhi Wickramasinghe, for their guidance and support throughout this project. Their feedback and knowledge greatly helped in the successful completion of this report.

~ END OF REPORT ~